# Practical 07 – Project

**In-Class Project: "CampusEats – Mini Food Ordering App"**

**Objective**
Design and implement a simple multi-screen Flutter application using:

- Column & Row
- Stack
- ListView.builder
- GridView
- Card & ListTile
- Navigator.push()
- setState() for cart count
- Expanded & Flexible for responsiveness

**Scenario**
University students often struggle to order food quickly during short breaks.
You are required to build a **simple prototype mobile app called "CampusEats"** for ordering food inside the university premises.
The app must contain **three screens only (simplified version for class)**:

1. Home Screen
2. Food Details Screen
3. Cart Screen

**Functional Requirements**
**1. Home Screen**
**Must Include:**

- AppBar with title **CampusEats**
- A GridView showing at least 6 food items
- Each food item displayed using:
  - Card
  - Image
  - Food name
  - Price
- A cart icon in AppBar showing number of selected items
- Clicking an item → navigate to Food Details screen

**Layout Concepts Tested:**

- GridView
- Column
- Card
- Expanded

- Navigator

**2. Food Details Screen**

**Must Include:**

- Large food image (use Stack to overlay discount badge)
- Food name
- Description
- Price
- "Add to Cart" button

**Functional:**

- When button is pressed:
  - Increase cart counter using setState
  - Show SnackBar confirmation

**Layout Concepts Tested:**

- Stack + Positioned
- Column
- ElevatedButton
- setState

**3. Cart Screen**

**Must Include:**

- ListView.builder showing selected items
- Each row:
  - Food name
  - Price
  - Remove button
- Total price at bottom
- Checkout button (no functionality required)

**Layout Concepts Tested:**

- ListView.builder
- Row
- Expanded
- setState for removal
- Basic calculation logic

**Minimum Data Structure (Students Must Create)**

```
class FoodItem {
  final String name;
  final double price;
  final String image;

  FoodItem({required this.name, required this.price, required this.image});
```

}

Students must create a List<FoodItem> with 6 items.

**UI Constraints**

Students must:

- Use at least one Stack
- Use Expanded or Flexible somewhere
- Use ListView.builder (not static ListView)
- Use Navigator.push()
- Use setState()